IKKISH

Arithmetic Problems

FOR BANKING, SSC & RAILWAY EXAM ASPIRANT

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Seven cousins named Aarav, Diya, Kabir, Mira, Riya, Vivek, and Zara are weighed one by one and the average weight increases by 4 kg every time. What is the difference between the weight of the lightest cousin and the heaviest cousin?

(A) 20 kg
(B) 28 kg
(C) 36 kg
(D) 44 kg
(E) None of these

In a container, the ratio of masala tea to herbal tea is 3 : 1. If 6X liters of this blend is removed and 9X liters of another blend that contains 80% herbal tea is mixed in, the final quantity becomes 60 liters more than the original. In the resulting mixture, the amount of masala tea exceeds the amount of herbal tea by 100 liters. What was the initial quantity of the blend?

(A) 524 liters
(B) 536 liters
(C) 548 liters
(D) 560 liters
(E) None of these

Ravi observes that the ratio of his boat's speed in still water to the speed of the current is 3 : 1. The boat requires the same amount of time to travel (N + 150) km downstream and (N – 250) km upstream. What is the value of N?

(A) 600
(B) 650
(C) 700
(D) 750
(E) None of these

A rhombus has a perimeter of 60 cm and one diagonal measures 18 cm. The area of a rectangle is 75% of the area of this rhombus. The length of the rectangle exceeds its breadth by 9 cm. Find the perimeter (in cm) of the rectangle.

(A) 72
(B) 62
(C) 54
(D) 80
(E) None of these

Suresh fills a container with 400 liters of a guava juice-water mixture, where guava juice constitutes 75% of the total. He then removes 100 liters of this mixture and adds 30 liters of guava juice along with 10 liters of water. What is the difference (in liters) between the amount of guava juice and water in the final mixture?

(A) 170
(B) 160

(C) 150
(D) 180
(E) None of these

Kirti has four ₹50 notes, three ₹100 notes, and five ₹200 notes in her purse. If she randomly picks three notes at the same time, in how many distinct ways can she get a total of exactly ₹350?

(A) 40
(B) 50
(C) 60
(D) 70
(E) None of these

Rahul beats Ajay by 80 m in a 400 m race. In an 800m race, Ajay beats Mehar by 30m or 7.5 seconds even when Ajay gave Mehar a head start of 20m. All three of them ran a race of ‘K’ m. They all started at the same time and Rahul completed the race in 60 seconds.

Quantity I: By how many meters did Rahul beat Mehar?

Quantity II: How many seconds did Mehar took to complete the race?

A. Quantity I < Quantity II

B. Quantity I ≤ Quantity II

C. Quantity I > Quantity II

D. Quantity I ≥ Quantity II

E. Quantity I = Quantity II or relationship can’t be established

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Twenty laborers were assigned to complete a job in a fixed number of days. Due to irregular attendance, only 7.50% of the work was finished in the first 15 days. From the 16th day onward, all 20 worked daily, yet only 45% of the total work got done by the deadline. How many days were originally allotted for the job?

(A) 20
(B) 24
(C) 28
(D) 30
(E) None of these

In a local election with three candidates—Sita, Geeta, and Reeta—Geeta secures 40% more votes than Sita. She also surpasses Reeta by 70,000 votes. Reeta obtains 30% fewer votes than Sita. If 80% of all registered voters cast valid ballots, how many people were on the voter list?

(A) 3,82,500
(B) 3,92,500
(C) 3,87,500
(D) 3,95,000
(E) None of these

Sunil departs from point P for point Q at a speed of (a + 2) km/hr, while Ramesh leaves point Q for point P at (a – 3) km/hr. Ramesh begins 2 hours earlier than Sunil, and they meet exactly in the middle of the 400 km route. How many hours would Sunil need to travel the entire distance from P to Q?

(A) 14
(B) 12.50
(C) 16
(D) 18
(E) None of these

A square sheet with a 28 cm side is joined along the shorter edge of a rectangular sheet measuring 60 cm in length and 28 cm in breadth. If the combined sheet is rolled along its total length to form a cylinder, what will be the radius of this cylinder?

(A) 12 cm
(B) 14 cm
(C) 21 cm
(D) 10.50 cm
(E) None of these

The HCF of 'A' and 50 is X and their LCM is 10X. The HCF of (A + 10) and 50 is X and their LCM is 150.

Quantity I: Find the value of A.

Quantity II: Find the value of X.

A. Quantity I < Quantity II

B. Quantity I ≤ Quantity II

C. Quantity I > Quantity II

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A vendor sets the marked price of an item 50% above its cost of Rs. 200. After two equal successive discounts of 'x'%, the item is sold at Rs. 192. If the vendor then raises the original marked price again by 'x'%, the new marked price will be what percent of the cost price?

(A) 150
(B) 160
(C) 180
(D) 200
(E) None of these

Suresh is 25% less efficient than Mahesh, and Suresh can finish a job in X days. If Suresh were 25% more efficient than Mahesh, he would complete the same job in X – 40 days. How many days would Mahesh alone require to complete the job?

(A) 60
(B) 75
(C) 80

(D) 85
(E) None of these

Ashok deposits Rs. $(2p + 3q)$ at 10% simple interest per annum for 2 years and finally gets Rs. $(4p + q)$. If $(p - q) = 2000$, find the difference between the compound interest at 15% per annum (compounded annually) for 2 years on the same principal and the simple interest he actually received.

(A) Rs. 2000
(B) Rs. 2250
(C) Rs. 2450
(D) Rs. 2750
(E) None of these

Ravi, Gita, and Mohan undertake a job together. In the first 8 days, all three complete 50% of the total work. Mohan then stops working, and the remaining part is split into two phases: Ravi and Gita finish 5/12 of the entire job in the next 12 days, after which Gita works alone for 4 more days to complete the task. Find the respective ratio of daily efficiencies of Ravi, Gita & Mohan.

(A) 2 : 4 : 3
(B) 3 : 2 : 4
(C) 4 : 2 : 3
(D) 2 : 3 : 4
(E) None of these

A container has 40 liters of milk-water mixture in the ratio 3 : 5. A certain amount of this mixture is removed and replaced with water. The new mixture's ratio of milk to water becomes 3 : 7. How many liters of the mixture were replaced?

(A) 6
(B) 7
(C) 8
(D) 9
(E) None of these

A and B started a business with investments of Rs. 6X and Rs. 5X, respectively. 2 years later, B left the business and at the same time C joined them with an investment of Rs. Y. Had B continued in the business, he would have received half as much profit as C after 5 years of business.

Quantity I: If A got a profit share of Rs. 1650, what was the profit share of C?

Quantity II: If B invested Rs. 960, how much did C invest?

A. Quantity I < Quantity II

B. Quantity I ≤ Quantity II

C. Quantity I > Quantity II

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Pankaj has Rs. 10 and Rs. 20 notes in a 3 : 4 ratio, and his total amount is Rs. 550. Manish holds Rs. 20 and Rs. 50 notes in a 1 : 1 ratio. If the number of Rs. 20 notes with Pankaj is exactly 5 more than the number of Rs. 50 notes with Manish, find how much money Manish has in total.
(A) Rs. 700
(B) Rs. 1050
(C) Rs. 900
(D) Rs. 1200
(E) None of these

Eight years ago, the ratio of Ramesh's age to his son Vikram's age was 10 : 1. If one-fourth of Ramesh's current age to twice Vikram's current age is in the ratio 1 : 2, find the difference (in years) between their present ages.

(A) 24
(B) 28
(C) 32
(D) 36
(E) None of these

Ramesh deposits a certain amount at 20% p.a. compound interest for 3 years. The interest of the 3rd year exceeds that of the 2nd year by Rs. 96. How much did he deposit initially?

(A) Rs. 1,500
(B) Rs. 1,800
(C) Rs. 2,000
(D) Rs. 2,500
(E) None of these

A shop owner buys a laptop for Rs. 200 and marks it at Rs. 300. By allowing a discount of p%, the final profit on the laptop is 12.5%. The same owner buys a smartphone for Rs. 400 and marks it at Rs. 600. Offering a discount of q% on the smartphone yields a 20% profit. Find p+q.

A) 20

B) 25

C) 35

D) 45

E) None of these

A boat makes a journey where it first goes ‘p’ km downstream and then returns to the starting point. Find the total time taken by the boat for the entire journey.

Statement I: The boat covered the last ‘0.25p’ km of the journey in 1.5 hours.

Statement II: The boat can cover the same journey in still water in 9.6 hours.

A. Statement I alone is sufficient, whereas Statement II alone is not sufficient.

B. Statement II alone is sufficient, whereas Statement I alone is not sufficient.

C. Either Statement I alone or Statement II alone is sufficient.

D. Both the statements together are necessary.

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Harsh and Suraj launch Business C, investing in the ratio 5 : 2. At the same time, Pallavi and Kunal start Business D with investments in the ratio 1 : 2. The combined investment of Suraj and Pallavi is 44.44% of the combined investment of Harsh and Kunal. If the profit from Business C is 50% of the profit from Business D, and their total combined profit is Rs. 6300, what is Kunal's share?

(A) Rs. 2100
(B) Rs. 2400
(C) Rs. 2800
(D) Rs. 3200
(E) None of these

The mean of 71 consecutive natural numbers is N. If the next 14 consecutive numbers are also included, by how much does the average increase?

(A) 4
(B) 5
(C) 6
(D) 7
(E) None of these

M is a natural number less than 100 and N is the number obtained on reversing its digits. What is the value of M/N?

Statement I: The difference between M and N is 9.

Statement II: M is divisible by 5.

A. Statement I alone is sufficient, whereas Statement II alone is not sufficient.

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Priya's salary became 2.16 times her original salary after two consecutive raises. If the second raise was 25% of the first (both in percentage terms), what was the percentage of the first raise?

(A) 66.67
(B) 40
(C) 50
(D) 80
(E) None of these

Vishal appears for an exam of five different subjects, each with maximum marks of 100, where he scores 32, p, q, r, and 88. His overall percentage score is 45%, and p, q, and r are distinct positive integers. Determine the maximum possible value of 2pq – r.

(A) 54119
(B) 54120
(C) 5402
(D) 5405
(E) None of these

Two vessels contain 30 liters and 45 liters of milk- water mixtures, respectively. In the first vessel, milk and water are in the ratio 1 : 2, while in the second vessel, they are in the ratio 2 : 1. Fifteen liters from the second vessel are poured into the first, and then nine liters of the resulting mixture in the first vessel are transferred back to the second. How many liters of milk does the second vessel finally contain?

(A) 20
(B) 22
(C) 24
(D) 26
(E) None of these

Ajay, Bhaskar, and Chetan invest Rs. 12,000, Rs. 16,000, and Rs. R respectively in a business. After 2 years, Ajay takes 20% of the total profit as the managing partner, and the remainder is shared according to their individual investments. In this distribution, Ajay and Bhaskar receive Rs. 5,400 and Rs. 3,600 respectively. If Ajay had not taken the 20% for managing, he and Bhaskar would have received Rs. 3,375 and Rs. 4,500 respectively. What is the value of R?

(A) 15,000
(B) 18,000
(C) 20,000
(D) 10,000
(E) None of these

The letters of the names "TEJAS" and "ANU" are combined to form T, E, J, A, S, A, N, U. In how many distinct ways can these 8 letters be arranged so that no two vowels are adjacent?

(A) 720
(B) 2880
(C) 840
(D) 1260
(E) None of these

Ishaan, Rahul, and Priya started a business investing 7, 9, and 8 units, respectively. After 6 months, Ishaan added 3 more units to his investment, Rahul added 1 more unit, while Priya withdrew 2 units. If the business ran for 1 year, in what ratio will their profits be divided?

(A) 17 : 19 : 18
(B) 19 : 17 : 15
(C) 17 : 19 : 14
(D) 16 : 19 : 16
(E) None of these

In Mixture P, the ratio of sugar to salt is 4 : 1, while in Mixture Q, the ratio of sugar to salt is 3 : 2. In what ratio should Mixture P and Mixture Q be mixed so that the resulting mixture has sugar to salt in the ratio of 7 : 3?

(A) 1 : 1
(B) 2 : 3
(C) 3 : 5
(D) 4 : 5
(E) None of these

Nitin sets aside 20% of his yearly salary. He places half of that amount in a 2-year compound interest plan at 10% per annum and the other half in a 2-year simple interest plan at the same rate. The difference between the compound interest and simple interest earned is Rs. 2000. What is Nitin's annual salary (in Rs.)?

(A) 10,00,000
(B) 24,00,000
(C) 15,00,000
(D) 20,00,000
(E) None of these

Rashi is currently 75% as old as Kunal. After some years, Rashi's age becomes 90% of Kunal's age at that time. By what percentage does Rashi's age increase during this period?

(A) 150%
(B) 200%
(C) 225%
(D) 250%
(E) None of these

5, 9, 7, X, 11, Y, 13, 15 is a sequence whose average is 10. If X is replaced by (4X+6) and Y is replaced by (Y+5), the average of the new sequence becomes 14. Find the value of X.

(A) 5
(B) 6
(C) 7
(D) 8
(E) None of these

Ravi and Sneha can complete a job in 10 days, whereas Sneha and Neha can finish the same job in 12 days. If Ravi's efficiency is twice as much as Neha's, in how many days will Sneha alone finish the job?

(A) 15
(B) 12
(C) 10
(D) 20
(E) None of these

Train G needs 8 hours more than Train Y to travel 600 km. If Train G's speed is doubled, it will take 2 hours less than Train Y to cover the same distance. What is the speed of Train Y?

(A) 40 km/h
(B) 45 km/h
(C) 50 km/h
(D) 55 km/h
(E) None of these

A bag contains some red, blue and green balls. Half of the balls are red and the probability of drawing a green ball is 1/6.

Quantity I: If the bag contains 120 balls, what will be the probability of drawing a blue ball after removing half of the green balls?

Quantity II: What will be the probability of drawing a red ball, after replacing 1/3rd of the red balls with green balls?

A. Quantity I < Quantity II

B. Quantity I ≤ Quantity II

C. Quantity I > Quantity II

D. Quantity I ≥ Quantity II

E. Quantity I = Quantity II or relationship can't be established

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On a straight route, City Nashik is located between Pune and Mumbai. Car P departs from Pune at 2 PM and Car Q leaves Nashik at the same time, both heading towards Mumbai. They arrive in Mumbai together at 6 PM. If Car P passes through Nashik exactly at 4 PM, what is the ratio of Car P's speed to Car Q's speed?

(A) 2 : 3
(B) 2 : 1
(C) 3 : 2
(D) 4 : 5
(E) None of these

Arjun and Virat purchase two items together for Rs. 10,000. Arjun's item is sold at a 25% profit, and Virat sells his item for Rs. 600 more than its cost. Together, they earn an overall profit of 20%. What is the selling price of Virat's item?

(A) Rs. 4000
(B) Rs. 5000
(C) Rs. 5400
(D) Rs. 5600
(E) None of these

----

a, b and c are three natural numbers such that a < b and c = 0.8b. The sum of the reciprocals of a and b is twice the difference between their reciprocals.

Quantity I: 0.75b × c

Quantity II: (3b + a)/a × c

A. Quantity I < Quantity II

B. Quantity I ≤ Quantity II

C. Quantity I > Quantity II

D. Quantity I ≥ Quantity II

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A. Quantity I < Quantity II

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Three cars belonging to Prakash, Qadir, and Rohan begin their journey from the same location along the same route. Prakash and Qadir set off at 2 pm, while Rohan departs at 4 pm. By 6 pm, Rohan catches up with Prakash and immediately increases his speed by 50%. With this new speed, Rohan manages to catch Qadir at 9 pm. What is the sum of the terms in the ratio of Prakash's speed to Qadir's speed?

(A) 16
(B) 18

(C) 20
(D) 22
(E) None of these

A reservoir of capacity 3600 cubic meters is connected to two pipes: an inlet pipe and an outlet pipe. The outlet pipe's flow rate surpasses the inlet pipe's flow rate by 15 cubic meters per minute. The outlet pipe empties the reservoir 20 minutes earlier than the inlet pipe takes to fill it. Find the inflow rate of the inlet pipe.

(A) 30 m^3/min
(B) 40 m^3/min
(C) 45 m^3/min
(D) 50 m^3/min
(E) None of these

A shopkeeper uses a faulty weight of 900 grams in place of 1000 grams while selling M kilograms of an item at the price of 30 kilograms, thereby gaining 50% profit. If the cost price is Rs. 20 per kg, what is the seller's price per kg?

(A) Rs. 20
(B) Rs. 24
(C) Rs. 27
(D) Rs. 30
(E) None of these

Train Jan-Shatabdi takes 10 seconds to pass a telegraph pole and 20 seconds to travel over a bridge that is 200 metres long. Another train, Rajdhani, is 150 metres longer than Jan-Shatabdi and moves at a speed 18 km/h faster. In how many second does Rajdhani crosses a 250m long platform?
(A) 28
(B) 18
(C) 24
(D) 32
(E) None of these

In a college, 70% of the students scored above 80% in their Class 10 boards, and among those, 2/7 secured more than 90%. The number of students scoring below 70% is 18, while the count of students who scored between 70% and 80% (inclusive) is one-seventh of those who crossed 80%. How many students in the college scored above 90%?

(A) 12
(B) 15
(C) 16
(D) 18
(E) None of these

Arjun has a bag containing 10 balls in total, some are red and the rest are green. The probability of drawing two red balls in succession without replacement is 2/9. How many red balls are in the bag?

(A) 3
(B) 4
(C) 5
(D) 6
(E) None of these

The length of a rectangle is equal to the diagonal of a square and the perimeter of the square is 6 times the breadth of the rectangle.

Quantity I: Find the area of the square.

Quantity II: Find the area of the rectangle.

A. Quantity I < Quantity II

B. Quantity I ≤ Quantity II

C. Quantity I > Quantity II

D. Quantity I ≥ Quantity II

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The length of a rectangle is equal to the diagonal of a square and the perimeter of the square is 6 times the breadth of the rectangle.

Quantity I: Find the area of the square.

Quantity II: Find the area of the rectangle.

A. Quantity I < Quantity II

B. Quantity I ≤ Quantity II

C. Quantity I > Quantity II

D. Quantity I ≥ Quantity II

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Karan can finish a job in 6 days, Nikhil in 12 days, and Mitesh in 4 days. All three started working together, but after some time Mitesh stopped. Karan and Nikhil then completed the remaining task in 2 days. After how many days did Mitesh leave?

(A) 1
(B) 2
(C) 3
(D) 4
(E) None of these

Rahul, Sunil, and Priya jointly invest in a new venture. Rahul puts in Rs. 375 for 8 months, Sunil invests Rs. 180 for 2 extra months compared to Rahul, and Priya invests Rs. 225 for the same duration as Sunil. If the combined profit share of Sunil and Priya is Rs. 350 more than the profit share of Rahul, find the profit share of Sunil.

A) Rs. 320

B) Rs. 750

C) Rs. 600

D) Rs. 450

E) None of these

Four numbers have an average of 14. The largest is 10 more than the smallest. The second largest is 4 more than the second smallest. The difference of the squares of the two smallest numbers is 105. Find the average of the squares of the second smallest and the second largest.

(A) 194
(B) 229
(C) 219
(D) 237
(E) None of these

The ratio of the salary of Amar and Prem is 4 : 9, respectively, and the salary of Karan is 37.50% of the average salary of Amar, Prem and Karan. Amar spends 75% of his salary and saves as much as Karan.

Quantity I: What is the percentage of salary spent by Karan?

Quantity II: Find the percentage savings of Orem, if his savings are twice the salary of Karan.

A. Quantity I < Quantity II

B. Quantity I ≤ Quantity II

C. Quantity I > Quantity II

D. Quantity I ≥ Quantity II

E. Quantity I = Quantity II or relationship can't be established

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The ratio of the salary of Amar and Prem is 4 : 9, respectively, and the salary of Karan is 37.50% of the average salary of Amar, Prem and Karan. Amar spends 75% of his salary and saves as much as Karan.

Quantity I: What is the percentage of salary spent by Karan?

Quantity II: Find the percentage savings of Prem, if his savings are twice the salary of Karan.

A. Quantity I < Quantity II

B. Quantity I ≤ Quantity II

C. Quantity I > Quantity II

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Manish and Suraj must complete a task, working on alternate days: Manish on day 1, Suraj on day 2, and so forth. The job finishes in 23 days. On day n, Manish completes n units of work, while Suraj's output remains the same each day. The ratio of Manish's first-day work to Suraj's second-day work is 1 : 3. How many days would Suraj alone need to complete the entire task?

(A) 35
(B) 47
(C) 59
(D) 62
(E) None of these

Bag X has 3 red, x green, and 8 yellow balls. The probability of drawing 2 green balls from Bag X is 1/12. Bag Y contains (x−1) white, (x+4) black, and (2x−3) orange balls. If 2 balls are drawn at random from Bag Y, in how many distinct ways can both be orange?

(A) 15
(B) 21
(C) 28
(D) 35
(E) None of these

Rohan arranges the digits 1 through 7 in a row. In how many distinct ways can he arrange them so that digits 3 and 6 always remain together?

(A) 720
(B) 1440
(C) 840

(D) 630
(E) None of these

Arun, Bhavesh, and Chirag invested Rs. (X + 600), Rs. X, and Rs. X in a venture for 16 months, 16 months, and 24 months respectively, and earned profits in the ratio 10 : 6 : 9. If instead they had invested Rs. 2X, Rs. X, and Rs. (X + 200) respectively for the same time periods, what would be Bhavesh's share from a total profit of Rs. 4350?

(A) Rs. 900
(B) Rs. 750
(C) Rs. 840
(D) Rs. 1050
(E) None of these

Mahesh spent a total of Rs. 23,000 on two types of items, P and Q. The ratio of the quantity of P to Q is 2 : 3, and each unit of Q costs 25% more than each unit of P. If Mahesh sells both types of items at a 50% profit, what is his overall profit percentage?

(A) 30
(B) 40
(C) 45
(D) 50
(E) None of these

Kunal, Rohan, and Shivam are three friends. When Rohan was half of his current age, he was 120% of what Shivam will be 6 years from now. When Rohan was 0.75 of his present age, Kunal was exactly 22 years old. Their present ages are in the ratio 5 : 9 : 3 (not necessarily in the order Kunal : Rohan : Shivam). What is the average of Rohan's and Shivam's current ages?

(A) 32 years
(B) 54 years
(C) 48 years
(D) 36 years
(E) None of these

A shopkeeper marked an item 'P'% above its cost price which is Rs. M. If the item is sold at a discount of Rs. '11.4P', he will get a profit of '0.4P'%.

Quantity I: What will be the percentage profit if the item is sold for Rs. 2470?

Quantity II: If selling the item at a discount of Rs. '0.17M' will earn the shopkeeper a profit of 15%, find the value of P.

A. Quantity I < Quantity II

B. Quantity I ≤ Quantity II

C. Quantity I > Quantity II

D. Quantity I ≥ Quantity II

E. Quantity I = Quantity II or relationship can't be established

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The average age of 15 colleagues is 30 years. Two people aged 70 and x leave the group. Three years later, a new colleague aged 66 joins. After 5 more years, the group's average age becomes 38. Find x.

A) 18

B) 20

C) 23

D) 25

E) None of these

After 3 years, the ratio of Niraj's age to Devika's age will be 4 : 5. Their current ages add up to 48. If Niraj's father, Suresh, is 27 years older than Niraj, find Suresh's present age.

(A) 50 years
(B) 46 years
(C) 51 years
(D) 48 years
(E) None of these

A, B and C are natural numbers such that the average of A and B is a 24 and the average of B and C is 29. The average of A, B and C is an integer. Neither A nor B is less than 23 or more than 27.

Quantity I: The average of a set of 7 natural numbers is 15. If C is also included in the set, what will be change in the average?

Quantity II: Find the difference between A and B.

A. Quantity I < Quantity II

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THANK - YOU